COMPUTER NETWORKS LABORATORY
COMPUTER NETWORK LAB TEST
EXPERIMENTS 1 AND 3

LAB TEST IMPLEMENTATION REPORT

Submitted by

Ms. UDAYA BHARGAVI RA2411030050033

Under the guidance of

Mrs. MALLIKKA RAJALINGAM

(Department of Cyber Security, Faculty of Engineering and Technology)

in partial fulfilment for the award of the degree

of

BACHELOR OF TECHNOLOGY

in

COMPUTER SCIENCE AND ENGINEERING w/s CYBER SECURITY

Of

FACULTY OF ENGINEERING AND TECHNOLOGY



SRM INSTITUTE OF SCIENCE AND TECHNOLOGY,
TIRUCHIRAPPALLI

OCTOBER 2026

EXPERIMENT 1

Introduction to Packet Tracer, Peer-to-Peer Communication, Study of Cables and its Color Codes

1.1 Aim

To become familiar with the Cisco Packet Tracer interface, to build a simple peer-to-peer network of two computers connected through a switch, to verify communication between them using the ping command, and to study the types of Ethernet cables and their colour codes.

1.2 Requirements

- Software: Cisco Packet Tracer
- Devices: two PCs (PC-PT) and one Catalyst 2960-24TT switch
- Cable: copper straight-through Ethernet cable

1.3 Procedure

1. Launch Cisco Packet Tracer and note the workspace, device-selection box and toolbar.
2. Place two PCs (PC0 and PC1) and one 2960-24TT switch (Switch0) on the workspace.
3. Connect each PC to the switch using a copper straight-through cable. The link lights turn green when the connection is up.
4. Open PC0, go to Desktop > IP Configuration and assign the IP address and subnet mask given in Table 1.1. Repeat for PC1.
5. Open the Command Prompt on PC0 and run ping 192.168.1.2 to test connectivity with PC1.
6. Observe the reply and the ping statistics.
7. Study the wire colour codes of the straight-through and crossover cables (Table 1.2).

Table 1.1: IP addressing

Device	IP Address	Subnet Mask
PC0	192.168.1.1	255.255.255.0
PC1	192.168.1.2	255.255.255.0

1.4 Network Topology

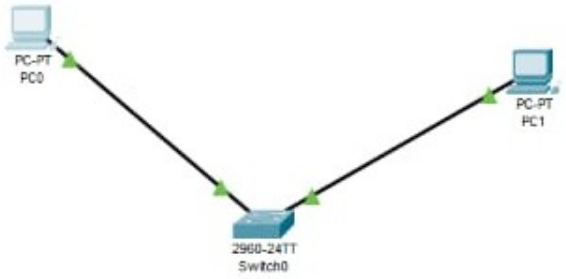


Figure 1.1: PC0 and PC1 connected through Switch0 (2960-24TT)

1.5 Output

The command ping 192.168.1.2 was executed from PC0. Four packets of 32 bytes were sent and four replies were received.

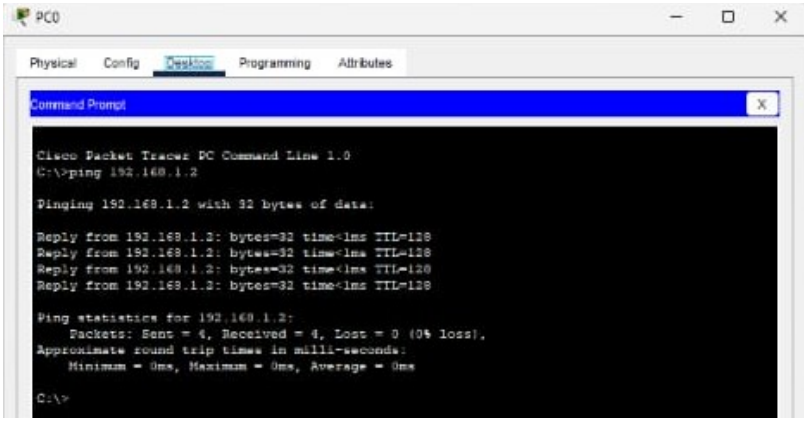


Figure 1.2: Ping from PC0 to PC1 (192.168.1.2)

Parameter	Observed value
Packets sent / received / lost	4 / 4 / 0 (0% loss)
Round-trip time	Minimum 0 ms, Maximum 0 ms, Average 0 ms
TTL	128
Bytes per packet	32

1.6 Study of Cables and Colour Codes

Ethernet cables contain four twisted pairs (eight wires) terminated in an RJ-45 connector. The wires are arranged according to the TIA/EIA-568 standard, which has two wiring schemes, T568A and T568B.

Table 1.2: Pin-wise colour codes

Pin	T568A	T568B
1	White/Green	White/Orange
2	Green	Orange
3	White/Orange	White/Green
4	Blue	Blue
5	White/Blue	White/Blue
6	Orange	Green
7	White/Brown	White/Brown
8	Brown	Brown

Table 1.3: Cable types

Cable	Wiring	Used to connect
Straight-through	Same standard (T568B) at both ends	Different devices, e.g. PC to switch, PC to router, switch to router
Crossover	T568A at one end and T568B at the other	Similar devices, e.g. PC to PC, switch to switch, router to router

1.7 Result

The two-PC network was created in Packet Tracer using a switch and straight-through cables. The ping from PC0 to PC1 succeeded with 0% packet loss, which verifies peer-to-peer communication, and the cable types and colour codes were studied.

EXPERIMENT 3

Router Configuration (Creating Passwords, Configuring Interfaces)

3.1 Aim

To configure a Cisco router by setting the console, enable and VTY passwords and by assigning IP addresses to its interfaces, and to verify communication between two networks through the router.

3.2 Requirements

- Software: Cisco Packet Tracer
- Devices: one Cisco 2911 router (Router0) and two PCs (PC2 and PC3)
- Cable: copper straight-through Ethernet cable

3.3 Procedure

1. Place one 2911 router and two PCs (PC2 and PC3) on the workspace and connect each PC to the router with a straight-through cable.
2. Click the router and open the CLI tab.
3. Enter privileged and global configuration mode and set the console, enable and VTY passwords.
4. Configure the interfaces GigabitEthernet0/0 and GigabitEthernet0/1 with IP addresses and enable them with no shutdown.
5. Assign the IP addresses and default gateways to PC2 and PC3 as given in Table 3.1.
6. Ping from PC2 to PC3 to test connectivity through the router.

Table 3.1: IP addressing

Device	Interface	IP Address	Default Gateway
Router0	Gig0/0	192.168.1.1/24	-
Router0	Gig0/1	192.168.2.1/24	-
PC2	FastEthernet0	192.168.1.2/24	192.168.1.1
PC3	FastEthernet0	192.168.2.2/24	192.168.2.1

3.4 Router Configuration Commands

```
Router> enable
Router# configure terminal
Router(config)# line console 0
Router(config-line)# password cisco
Router(config-line)# login
Router(config-line)# exit
Router(config)# enable password cisco
Router(config)# line vty 0 4
Router(config-line)# password cisco
Router(config-line)# login
Router(config-line)# exit
Router(config)# interface gigabitEthernet 0/0
Router(config-if)# ip address 192.168.1.1 255.255.255.0
Router(config-if)# no shutdown
Router(config-if)# exit
Router(config)# interface gigabitEthernet 0/1
Router(config-if)# ip address 192.168.2.1 255.255.255.0
Router(config-if)# no shutdown
Router(config-if)# end
```

3.5 Network Topology

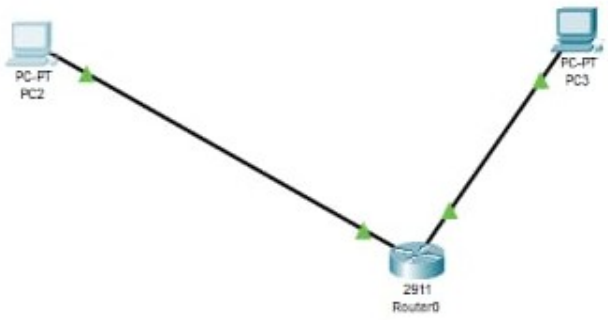


Figure 3.1: PC2 and PC3 connected to Router0 (2911)

3.6 Output

The command ping 192.168.2.2 was executed from PC2. Three of the four packets received replies.

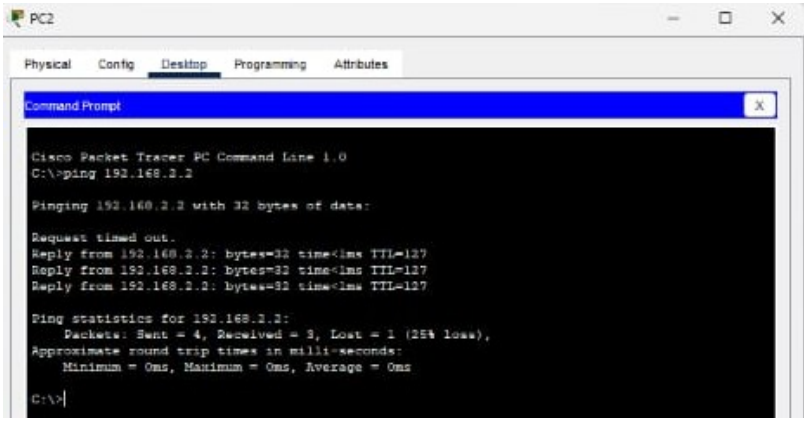


Figure 3.2: Ping from PC2 to PC3 (192.168.2.2) through the router

Parameter	Observed value
Packets sent / received / lost	4 / 3 / 1 (25% loss)
Round-trip time	Minimum 0 ms, Maximum 0 ms, Average 0 ms
TTL	127
Bytes per packet	32

3.7 Observation

The first request timed out because the PC and the router had to resolve each other's MAC addresses through ARP before the first packet could be delivered. The remaining three packets were answered normally. The TTL of 127 is one less than the default value of 128 because the packets crossed one router hop, which confirms that the traffic passed through Router0.

3.8 Result

The router was configured with console, enable and VTY passwords and with IP addresses on both interfaces. PC2 in the 192.168.1.0/24 network successfully communicated with PC3 in the 192.168.2.0/24 network through the router.